

Week 2 lecture 1

“Lexical elements” (MATLAB API)

MATLAB API

Commonalities



- ◆ Operators
- ◆ Variables
- ◆ Functions
- ◆ Keywords
- ◆ Types (again)

Operators

- ◆ Any calculation of homogeneous type descriptor
- ◆ Addition takes some amount of numbers and returns a number

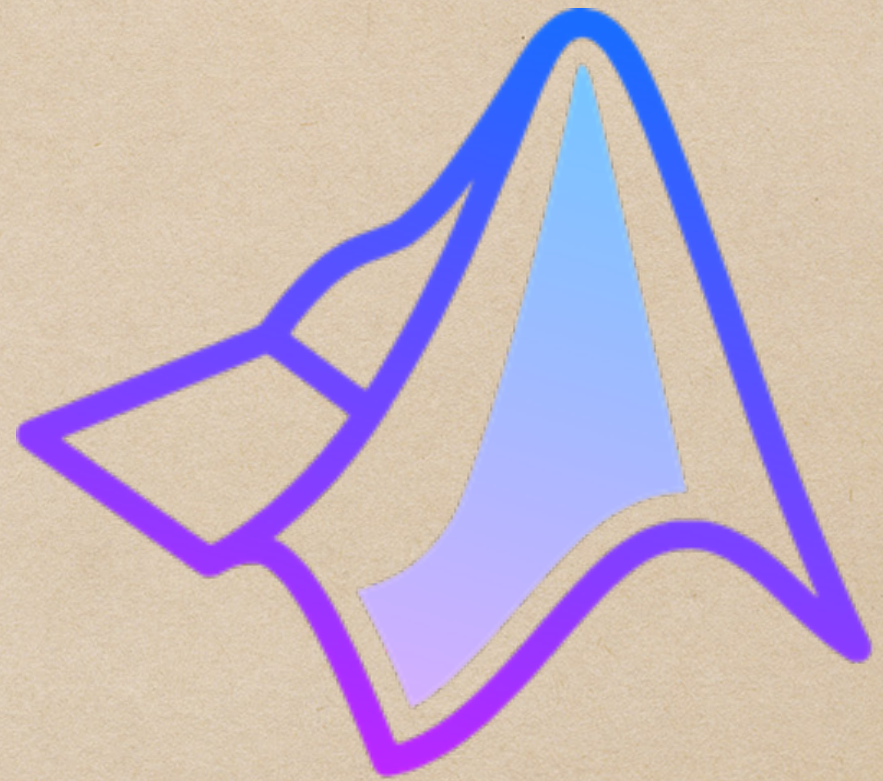
Common Operators

- Association: ()
- Unary: positive +, negative -
- Arithmetic: +, -, *, /, ^ log, \%
- Relational: <, >, <=, >=, ==, ~=
- Logical (Bitwise): &, &&, |, ||

Operators: Why?

- ◆ Custom Operators
- ◆ Function creation
- ◆ Modular sequencing of algorithms
- ◆ Abstraction

Variables



MATLAB®

Variables in MATLAB

- ◆ Include Type
- ◆ Stored in memory
- ◆ Lookup table

Variables: Common methods

- ◆ *who*: list of variables
- ◆ *whos*: detailed variables table

Variables: theory vs MATLAB

Theory

MATLAB

Var: Name x Value

Var: Name x Value x Type

$e \leftarrow (20 + 5)6$

$A = \sin(60)$

Collections



Theory of Collections

- ◆ Homogeneous
- ◆ Bag of things

Collection: Array

- ◆ Ordered (indexable)
- ◆ Finite

Collections: Vectors

- ◆ Ordered
- ◆ Dynamic
- ◆ Directional

Linear Algebra

(Vector) Matrix

- ◆ Utilizes Vectors
- ◆ Multi-dimensional
- ◆ Theory based on Linear Algebra